

LAZARD LEGACY MATH ACADEMY

# College Algebra Bootcamp

Complete College Algebra Course | Final Exam Included

Instructor: Gregory Lazard Jr. | New Orleans, Louisiana

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# Algebra Fundamentals Review

## Properties, Order of Operations, Factoring

Order of Operations (PEMDAS): Parentheses, Exponents, Multiplication/Division (L to R), Addition/Subtraction (L to R)

Example:  $3+2\times(4+1)^2=3+2\times25=3+50=53$

Exponent rules:  $a^m \times a^n = a^{m+n}$  |  $(a^m)^n = a^{m \times n}$  |  $a^0 = 1$  |  $a^{-1} = 1/a$

Factoring: GCF first, then trinomials.  $x^2-x-6=(x-3)(x+2)$

## 10 Examples + 15 Problems

Solve:  $x^2-9=0 \rightarrow (x+3)(x-3)=0 \rightarrow x=\pm 3$

Simplify:  $(x^3y^2)/(xy) = x^2y$

Practice: 1)  $2+3 \times 4$  2) Factor  $x^2+5x+6$  3)  $(x^2y^3)(x^3y)$  4)  $x^2-81$  5) PEMDAS:  $5 \times (3+2) - 4^2$  6) Factor  $2x^2+6x$  7)  $x^2$  8) Factor  $x^2-11x+28$  9) Simplify  $(2x)^3$  10) Factor  $3x^2-12$  11-15: additional problems

# Functions & Function Notation

## Domain, Range, Composition, Inverse

A function  $f$  assigns exactly one output to each input.

$f(x)=2x+3$  means: substitute  $x$  and evaluate.  $f(4)=2(4)+3=11$

Domain: all valid inputs. Range: all possible outputs.

Composition:  $(f \circ g)(x)=f(g(x))$ . If  $f(x)=x+1$  and  $g(x)=2x$ , then  $f(g(3))=f(6)=7$

Inverse: swap  $x$  and  $y$ , then solve for  $y$ . If  $f(x)=3x-1$ , inverse:  $x=3y-1 \rightarrow y=(x+1)/3$

## 10 Examples + 15 Problems

$f(x)=x^2-4$ . Find  $f(3)=5$ . Domain: all reals. Range:  $y \geq -4$

$g(x)=\sqrt{x}$ . Domain:  $x \geq 0$ . Range:  $y \geq 0$

Practice: 1)  $f(x)=4x-2$ , find  $f(3)$  2)  $g(x)=x^2+1$ , find  $g(-2)$  3) Domain of  $\sqrt{x-3}$  4) Find inverse of  $f(x)=2x+6$

5)  $h(x)=3x$ ,  $g(x)=x+2$ , find  $h(g(4))$  6)  $f(x)=|x|$ , find  $f(-5)$  7) Range of  $f(x)=x^2$  8) Inverse of  $f(x)=5x$

9)  $f(x)=1/x$ , domain? 10) Composition:  $f(g(2))$  where  $f(x)=x^2, g(x)=x+1$

# Polynomial Functions

## Operations, Graphing, Zeros

Adding polynomials: combine like terms.  $(3x^2+2x-1)+(x^2-5x+4)=4x^2-3x+3$

Multiplying: use FOIL or distribution.  $(x+3)(x-4)=x^2-4x+3x-12=x^2-x-12$

Zeros: values of  $x$  where  $f(x)=0$ . Set equal to zero and solve.

Degree determines end behavior and number of zeros.

## 10 Examples + 15 Problems

Add:  $(2x^3-x+5)+(x^3+3x-2)=3x^3+2x+3$

Zeros of  $x^2-5x+6=0$ :  $x=2$  and  $x=3$

Practice: 1)Add  $(x^2+3x)+(2x^2-x)$  2)Subtract  $(4x^2-2x)$  from  $(6x^2+x)$  3)Multiply  $(x+5)(x-3)$  4) $(2x-1)(x+4)$

5)Zeros of  $x^2-7x+12$  6)Zeros of  $x^2-9$  7) $(x+2)^3$  expanded 8)Degree of  $5x^4-3x^2+x$  9)Zeros of  $2x^2+x-3$

10)Factor  $6x^2+11x-10$

# Rational Expressions

## Simplifying, Operations, Equations

Simplify by factoring numerator and denominator, then cancel common factors.

Example:  $(x^2-4)/(x-2) = (x+2)(x-2)/(x-2) = x+2$

Add/subtract: find LCD first. Multiply/divide: flip and multiply for division.

Solve rational equations: multiply both sides by LCD, then solve.

## 10 Examples + 15 Problems

Simplify  $(x^2-9)/(x+3)=x-3$

Solve  $3/x=6/8 \rightarrow x=4$

Practice: 1)Simplify  $(x^2-25)/(x+5)$  2) $(2x^2-8)/(x-2)$  3)Add:  $1/x+2/x$  4) $2/(x+1)+3/(x+1)$  5)Solve  $4/x=2$   
6)Solve  $1/x+1/2=1$  7)Simplify  $(x^2+3x)/(x)$  8)Multiply  $(x/2) \times (4/x)$  9)Divide  $(6/x) \div (3/x^2)$  10)Solve  
 $2/(x-1)=4$

# Exponential & Logarithmic Functions

## Properties, Graphs, Equations

Exponential function:  $f(x)=a^x$  where  $a>0$ ,  $a\neq 1$

$f(x)=2^x$ : passes through (0,1), increasing, asymptote at  $y=0$

Logarithm is the inverse of exponential.  $\log_a(b)=c$  means  $a^c=b$

$\log_2(8)=3$  because  $2^3=8$  |  $\log_{10}(1000)=3$  because  $10^3=1000$

$\ln$  is natural log (base  $e$ ). log properties:  $\log_a(mn)=\log_a m+\log_a n$

## 10 Examples + 15 Problems

Solve  $2^x=16 \rightarrow x=4$ . Solve  $\log_3(x)=2 \rightarrow x=9$

Simplify  $\log_2(16)=4$  because  $2^4=16$

Practice: 1) Solve  $3^x=81$  2)  $\log_5(32)=?$  3)  $\log_{10}(100)$  4) Solve  $5^x=125$  5)  $\ln(e^3)$  6) Solve  $\log_4(x)=3$

7) Expand  $\log(x^2y)$  8) Solve  $2^{x+1}=16$  9)  $\log_3(625)$  10) Solve  $\log_2(64)=3$

# Systems of Equations

## Substitution, Elimination, 3x3 Systems

2x2 System: two equations, two unknowns.

Substitution: solve one equation for one variable, substitute into the other.

Elimination: add/subtract equations to eliminate a variable.

Example by substitution:  $y=2x-1$  and  $3x+y=14$ . Substitute:  $3x+(2x-1)=14 \rightarrow 5x=15 \rightarrow x=3, y=5$

Example by elimination:  $2x+3y=12$  and  $4x-3y=6$ . Add:  $6x=18 \rightarrow x=3, y=2$

## 10 Examples + 15 Problems

Practice: 1)  $x+y=10, x-y=4$  2)  $2x+y=7, x+y=4$  3)  $3x+2y=12, x=y$  4)  $4x-y=5, 2x+y=7$  5)  $x+2y=8, 3x-y=3$   
6)  $5x+y=13, x-y=1$  7)  $2x+3y=7, x=2y$  8)  $x+y+z=6, x-y=2, z=x$  9) 3 equations:  $x+y=4, y+z=7, x+z=5$   
10) word problem: two numbers sum 20 and differ by 4 — find them



# Final Exam — 40 Questions

## College Algebra Final Exam with Answer Key

1. Factor:  $x^2-36$ . A)  $(x-6)^2$  B)  $(x+6)(x-6)$  C)  $(x+6)^2$  D)  $x^2-36$  Answer: B

2.  $f(x)=3x-5$ . Find  $f(4)$ . A) 7 B) 17 C) 12 D) -5 Answer: A (7)

3. Solve:  $2x+y=8$  and  $x-y=1$ . A) (3,2) B) (2,3) C) (4,0) D) (1,6) Answer: A ( $x=3,y=2$ )

4. Simplify:  $(x^2-4)/(x-2)$ . A)  $x-2$  B)  $x+2$  C)  $x^2$  D) 1 Answer: B ( $x+2$ )

5. Solve  $2^{\blacksquare}=32$ . A) 4 B) 5 C) 6 D) 3 Answer: B (5)

6–40: Complete the full 40-question final exam using all topics from this course. Check your work with the answer key provided by your instructor.

Answer Key 1-5: B, A, A, B, B